

Question	Answer	Marks
7(a)	number of protons = 95	A1
	number of neutrons = 146	A1
7(b)	Np/neptunium (nucleus) has <u>kinetic</u> energy or gamma/ γ -radiation produced	B1
7(c)(i)	$I = NQ / t$	C1
	$I = (6.9 \times 10^{11} \times 2 \times 1.60 \times 10^{-19}) / 30$	A1
	$= 7.4 \times 10^{-9} \text{ A}$	
7(c)(ii)	$P = (6.9 \times 10^{11} \times 5.5 \times 10^6 \times 1.60 \times 10^{-19}) / 30$	C1
	$= 0.020 \text{ W}$	A1